

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA130 |

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

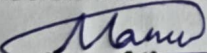
Total Marks:25

Analytical Problem

Code of Conduct:

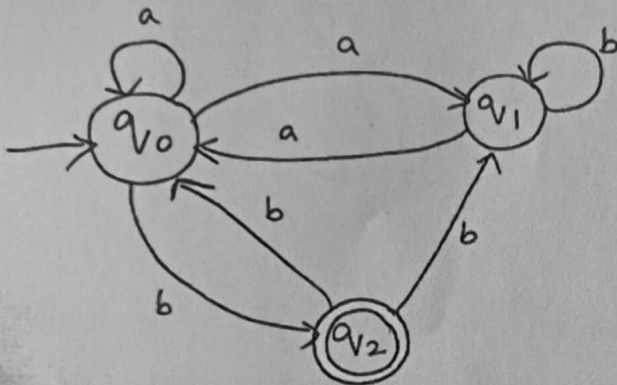
I C. Manohar / 192372246 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L

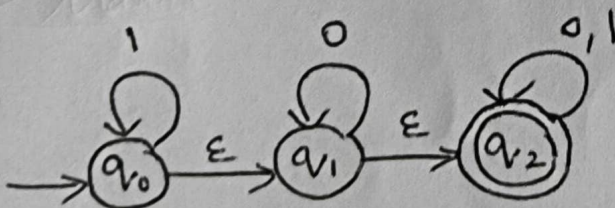


2. Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb , Construct an NFA for L , Then draw the equivalent DFA

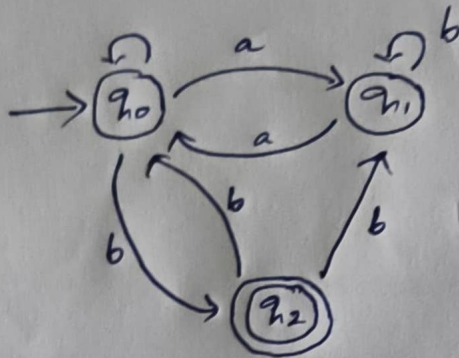
3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



① Given state diagram:



Step: 2 - Transition Table

state/ip	a	b
q ₀	{q ₀ , q ₁ }	{q ₂ }
q ₁	{q ₀ }	{q ₁ }
q ₂	{ $\emptyset$ }	{q ₀ , q ₁ }

Step 3:

Formula: $\{\emptyset, \Sigma, \delta, q_0, F\}$

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{a, b\}$$

$$\delta(q_0, a) = \{q_0, q_1\}$$

$$\delta(q_0, b) = \{q_2\}$$

$$\delta(q_0, a) = \{q_0\}$$

$$\delta(q_0, b) = \{q_1\}$$

$$\delta(q_2, a) = \emptyset$$

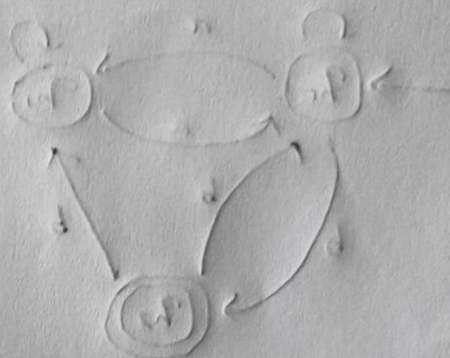
$$\delta(q_2, b) = \{q_0, q_1\}$$

Step 4:

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_0, q_1\}$$

$$\epsilon\text{-closure}(q_2) = \{q_0, q_1, q_2\}$$



Steps:

$$\begin{aligned}
 \delta'(q_0, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, a), a)) \\
 &= \epsilon\text{-closure}(\delta(\{q_0, q_1, q_2\}, a)) \\
 &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\
 &= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \{\emptyset\}) \\
 &= \epsilon\text{-closure}(q_1 \cup q_0) \\
 &= \epsilon\text{-closure}(q_0, q_1) \cup (q_0, q_1, q_2) \\
 &= \epsilon\text{-clos. } \{q_0, q_1, q_2\}
 \end{aligned}$$

$$\begin{aligned}
 \delta'(q_0, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), b)) \\
 &= \epsilon\text{-closure}(\delta(\{q_0, q_1, q_2\}, b)) \\
 &= \epsilon\text{-closure}(q_2 \cup q_1, \{q_0, q_1\}) \\
 &= \{q_0, q_1, q_2\}
 \end{aligned}$$

$$\delta'(q_1, a) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), a))$$

$$= \epsilon\text{-closure}(\delta(q_0, q_1), a)$$

$$= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a))$$

$$= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\})$$

$$= \epsilon\text{-closure}(\{q_0, q_1\})$$

$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_1, b) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), b))$$

$$= \epsilon\text{-closure}(\delta(q_0, q_1), b)$$

$$= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b))$$

$$= \epsilon\text{-closure}(q_2 \cup q_1)$$

$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_2, a) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), a))$$

$$= \epsilon\text{-closure}(q_0, q_1, q_2, a)$$

$$= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a))$$

$$= \epsilon\text{-closure}(\{q_0 \cup q_1\})$$

$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_2, b) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), b))$$

$$= \epsilon\text{-closure}(\delta(q_0, q_1, q_2), b)$$

$$= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b))$$

$$= \epsilon\text{-closure}(q_2 \cup q_1 \cup \{q_0, q_1\})$$

$$= \{q_0, q_1, q_2\}$$

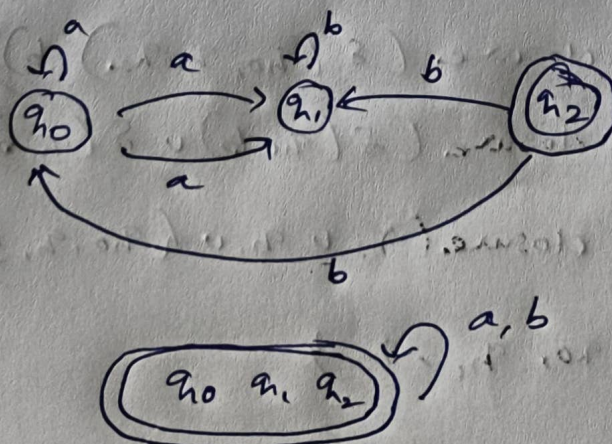
$$\begin{aligned}
 \delta'(q_0, q_1, q_2, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), a)) \\
 &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\
 &= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \emptyset) \\
 &= \{q_0, q_1, q_2\}
 \end{aligned}$$

$$\begin{aligned}
 \delta'(q_0, q_1, q_2, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), b)) \\
 &= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b)) \\
 &= \epsilon\text{-closure}(\emptyset \cup \{q_1\} \cup \{q_0, q_1\}) \\
 &= \{q_0, q_1, q_2\}
 \end{aligned}$$

Transition Table:

State/p	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	$\emptyset$	$\{q_0, q_1\}$
q_0, q_1, q_2	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$

NFA State diagram:



Language:

strings of length atleast 2 whose last two symbols are different

Ex:

Accepted

ab

ba

aaab

abaaab

Rejected

bb

a

b

NFA:

States

q_0 = start

q_1 = last symbol a

q_2 = last symbol b

q_3 = final

Transitions:

$q_0 \xrightarrow{a} q_1$

$q_0 \xrightarrow{b} q_2$

$q_1 \xrightarrow{a} q_1$

$q_1 \xrightarrow{b} q_3$

$q_2 \xrightarrow{b} q_2$

$q_2 \xrightarrow{a} q_3$

$q_3 \xrightarrow{a} q_1$

$q_3 \xrightarrow{b} q_2$

Final state:

q_3

Equivalent DFA:

S_0 = start

S_1 = Ends with a

S_2 = Ends with b

S_3 = Ends with ab (Accept)

S_4 = Ends with ba (Accept)

Transition Table:

State	a	b
S_0	S_1	S_2
S_1	S_1	S_3
S_2	S_4	S_2
S_3	S_4	S_2
S_4	S_1	S_3

Final state:

S_3 S_4

Language:

Ends with

00 (02) 11

Formal definition:

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

$$\text{Start} = q_0$$

$$\text{Final} = \{q_3, q_4\}$$

Transition function:

State	0	1
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_3\}$	$\emptyset$
q_2	$\emptyset$	$\{q_4\}$
q_3	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$(\emptyset)$

String:

01011

$q_0 \rightarrow 0$ $q_0, q_1 \rightarrow 1$ $q_0, q_2 \rightarrow 0$ $q_0, q_3 \rightarrow 1$

$q_0, q_4 \rightarrow 1$ q_4

Since q_4 is final

$\therefore 01011$ is Accepted

4. Language:

$L = \{011\}$

Need substring:

001

States:

$q_0 = \text{No Match}$

$q_1 = \text{Seen } 0$

$q_2 = \text{Seen } 00$

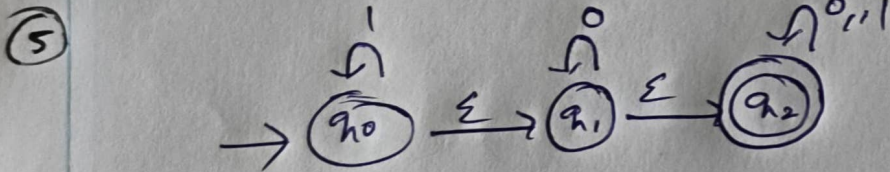
$q_3 = \text{Seen } 001 \text{ (Final)}$

Transition Table:

State	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_4

Final state:

q_3 this DFA accepts every string containing 001.



Given:

$q_0 \xrightarrow{0} q_1$

$q_1 \xrightarrow{0} q_2$

q_0 loop on 1

Final:

q_2

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

Start = q_0

Final : $\{q_3, q_4\}$

Transition Function:

State

0

1

A

B

A

B

C

C

C

C

C

DFA State:

$$A = \{q_0, q_1, q_2\}$$

$$B = \{q_1, q_2\}$$

$$C = \{q_2\}$$

Start state:

A

Final state:

A B C

